Scaling Up Memory Disaggregated Applications with Smart

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1 Background

1.1 RDMA Netowrk

RDMA supports one-sided verbs, i.e., READ, WRITE, CAS (compare-and-swap), and FAA (fetch-and-add), which operate directly on remote memory without involving weak CPUs in memory blades.

The programming interface of RDMA is centralized around the abstraction of queue pairs (QPs). Each QP contains a pair of send queue and receive queue and is associated with a completion queue (CQ). For each RDMA operation, the library posts a work request (WR) to the QP. Then, after a WR is acknowledged by the receiver's RNIC, the sender's RNIC performs a PCIe DMA write that appends a CQ entry, and the application is notified by continuously polling the CQ.

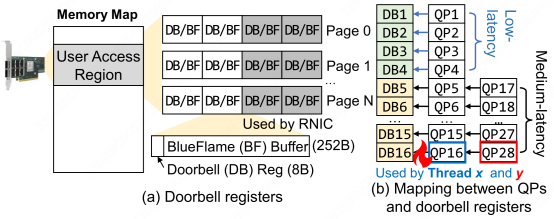

Figure 2. Doorbell registers in Mellanox ConnectX-6.

Figure 1

Doorbell Register (DB) is the mechanism used by the RNIC to receive notifications of newly enqueued work requests. Figure 1 illustrates the doorbell registers in ConnectX-6. By default, up to 16 DBs are allocated per

RDMA device context, including 4 low-latency DBs and 12 medium-latency DBs. Each low-latency DB is associated with only one QP, but multiple QPs can be associated with the same medium-latency DB. These doorbell registers are mapped in the User Access Region (UAR), and the driver library performs memory-mapped I/O (MMIO) writes to update them. For drivers used

by Connect-IB to ConnectX-7 , updates to the same DB are all protected by a spinlock[1] . As a result, when a DB is shared by multiple QPs used by different threads, an implicit thread contention occurs and becomes an implicit bottleneck that limits the parallelism of disaggregated applications. For ex- ample, as shown in Figure 2(b), thread □ using QP16 may be contended by thread □ using QP28 because both QPs are associated with DB16. Obviously, such a default mapping is not optimized for massively parallel applications. Memory Translation and Protection Table. Many existing works worry that too many QPs will exhaust the small on-chip SRAM cache in the RNIC and lead to cache thrash- ing problems [5, 16, 25, 41]. But, the caches inside RNICs are actually used to cache many other important objects whose sizes are not proportional to the number of QPs [26]. For example, once the RNIC is notified by the doorbell ringing,

2 Problems

3 References

[1]Enabling the `MLX5_SINGLE_THREADED` environment variable removes such spinlocks, but leads potential races between two CPUs due to write-combining buffers in multi-threaded applications